

IAT461 – Progress Report: BC Hydro EV Charging & Grid Load Optimization

Project: BC Hydro EV Charging & Grid Load Optimization

Data Scientist: Tim Kung | **Client:** Phumnawat (Poom) Phosawatmanee | **Date:** August 5, 2026

1. Project & Phase Alignment

The work completed to date corresponds to Phase 1: EDA & Strategy Validation as defined in the client proposal. This phase requires analyzing the EV charging sessions dataset to evaluate the three candidate policy responses (Option A - peak tariff penalty, Option B - transformer capital upgrade, Option C - clustering-based targeted incentives), building the case for why a data-science-driven segmentation may be the appropriate path forward once Options A and B are evaluated against the available data.

Cross-referencing this against our lab agreement, Phase 1 maps directly onto the Week 1 milestone: "EDA & first clustering pass." Our agreement specifies that Week 1 should deliver a neutral, evidence-based evaluation of Options A/B/C; explicit documentation of the Option A equity argument as a stated client assumption rather than a data-derived finding; feature engineering (cyclical time encoding, energy-to-duration ratio, peak-hour overlap, scaling); inclusion of housing_type directly as a clustering input; and a first-pass comparison of K-Means and DBSCAN using Silhouette Score. All five of these deliverables are present in the current notebook, indicating the project is tracking on schedule against the agreed 3-week plan.

2. Exploratory Data Analysis (EDA)

Data Overview & Preprocessing

- Three source tables were loaded: sessions (1,200 EV charging records, 10 fields), consumption (48 monthly BC-wide MWh readings), and authority_load (8,760 hourly BC Hydro control-area load readings).
- start_time was converted to a proper datetime and used to derive day_of_week, month, and a 30-minute time-of-day bucket for demand-profile analysis.
- Total energy usage was aggregated by month (relatively flat, ~2,000–2,900 kWh/month) and by day of week (weekday range ~3,800–4,800 kWh, no strong weekly seasonality).

Evidence Review of Client Options

Option A (peak tariff, 4–9 PM):

- peak_overlap_ratio shows a quarter of sessions never touch the surcharge window (25th percentile = 0), median 0.29. Overall, ~30% of total session energy falls inside the 4–9 PM window, and exposure is fairly uniform across housing types (Condo/Apartment 32.3%, Townhouse 29.4%, Single-Family 29.1%) — no dwelling type is disproportionately exposed by volume. The equity concern (that the surcharge penalizes non-EV, low-income households) was explicitly flagged as a stated client assumption, not a data-backed finding, since the dataset lacks income and household_EV_ownership fields.

Option B (transformer upgrade):

- Single-Family sessions carry the largest share of total charging energy (50.1%, n=600), proportional to their share of sessions overall. However, the dataset has no neighbourhood or transformer identifier, and authority_load.csv only reports demand

at the whole-of-territory level — so the analysis cannot localize where transformer stress actually concentrates, which is the central question a capital upgrade decision needs answered.

Option C (clustering-based incentives):

- Session-level behaviour is highly heterogeneous. `duration_hours` is bimodal (short Level-2 top-ups vs. long overnight Level-1 sessions), `peak_overlap_ratio` spans the full 0–1 range, and `battery_soc_start` is roughly uniform. This spread supports the premise that distinguishable charging-habit groups exist and may be worth targeting.

Feature Engineering & Key Data-Quality Findings

- Charger power parsed to kW, cyclical (sin/cos) encoding of start hour, an `is_weekend` flag, one-hot encoding of `housing_type` and season, and an `energy_to_duration_ratio` (delivery rate).
- A redundancy check revealed `energy_kwh` and `battery_soc_start` are correlated at exactly -1.00, which is likely a synthetic-data artifact rather than an empirical relationship. `energy_kwh` was excluded from the clustering feature matrix to avoid double-counting this signal, while retained separately for cluster profiling in familiar kWh terms.

Clustering Results

- The four K-Means clusters (n=180, 429, 343, 248) are separated primarily by charger level and peak-hour timing rather than battery state-of-charge
- DBSCAN (eps=1.8, min_samples=15, chosen via k-distance plot) found only 2 clusters and 142 noise points (~11.8%), essentially recovering the binary Level 1 vs. Level 2 charger-hardware split
- Both methods agree that `housing_type` and season do not meaningfully differentiate the clusters, reinforcing that this is a behavioural, not demographic, segmentation, thus directly satisfying the client's requirement that clusters be validated for interpretability, not just Silhouette Score.

Preliminary risks

- The dataset's lack of `neighbourhood`, `transformer_id`, and `income/household_EV_ownership` fields limits Options A and B to proxy-level evidence (`housing_type`, aggregate energy share). This should be flagged to the client as a scope limitation rather than resolved within the current data.
- The `energy_kwh` / `battery_soc_start` redundancy required an unplanned mid-EDA adjustment to the feature set, which changed the optimal cluster count from 5 to 4. This is a reasonable, well-documented correction but represents a scope deviation worth noting.
- PCA visualization retains only 47.7% of total variance (PC1 28.4%, PC2 19.3%), so 2D cluster plots should be presented as illustrative rather than a complete picture of separation.
- DBSCAN's eps parameter lacked a sharp elbow due to the near-binary `charger_power_kw` feature, requiring a manual noise/purity trade-off; this subjectivity should be noted transparently in the final write-up.